

Report on PDE-Transformer: Efficient and Versatile Transformers for Physics Simulations

SAMET KOGBAY, Technical University of Munich

1 INTRODUCTION

Foundation models have achieved remarkable success in computer vision and natural language processing, yet their adoption in scientific machine learning particularly for Partial Differential Equations (PDEs) has been limited. This gap arises from the multi-modal and multi-scale nature of PDE data, which varies across physical channels, boundary conditions, and spatial resolutions. Such variability makes it difficult to directly apply standard foundation-model architectures to physics simulations.

This report summarizes “PDE-Transformer: Efficient and Versatile Transformers for Physics Simulations” [Holzschuh et al. 2025]. The authors identify key shortcomings of existing transformer architectures, including Vision Transformers (ViTs) [Dosovitskiy et al. 2021] and Diffusion Transformers (DiTs) [Peebles and Xie 2023]. In particular, global self-attention leads to prohibitive quadratic scaling at high resolutions, while fixed channel embeddings hinder generalization across different physical systems.

To address these challenges, the authors propose the PDE-Transformer, a transformer backbone explicitly designed as a foundation model for physics simulations on regular 2D grids. The architecture extends DiT with a U-shaped multi-scale design and a novel Separate Channel embedding strategy, enabling efficient scaling and improved generalization across diverse PDEs. By supporting both deterministic surrogate modeling and probabilistic generation via flow matching, PDE-Transformer positions itself as a strong candidate for a general-purpose physics foundation model.

2 BACKGROUND: TRANSFORMERS AND THE SCALING BOTTLENECK

The standard Transformer architecture relies on the global self-attention mechanism to capture long-range dependencies in data. Given an input sequence of N tokens, the attention operation computes pairwise interactions between every token, resulting in a computational and memory complexity of $O(N^2)$ [Vaswani et al. 2017].

This quadratic scaling is manageable for many language and low-resolution vision tasks, but it becomes a severe bottleneck for physics simulations. Physics data is typically represented on high-resolution grids. For instance, a 2D simulation grid of size $H \times W$ results in $N = H \cdot W$ tokens. As the resolution increases to capture fine-scale physical phenomena (e.g. turbulence), N grows effectively quadratically, causing the cost of global attention (N^2) to explode.

This scaling issue renders standard ViTs and DiTs computationally prohibitive for high-fidelity surrogates, necessitating the efficient attention mechanisms discussed in the following methodology.

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The original work is introduced by [Holzschuh et al. 2025].

3 ADAPTING TRANSFORMERS FOR PHYSICS

To address the limitations of standard Vision Transformers in scientific computing, the authors propose the PDE-Transformer. This architecture modifies the DiT backbone to accommodate the multi-scale, high-resolution, and heterogeneous nature of physics simulations. It is important to note that unlike DiTs which operate in latent space the PDE-Transformer operates on raw data.

3.1 Backbone Architecture: U-Shaped Multi-Scale Design

Standard DiT architectures typically operate on a fixed latent dimension, whereas physics simulations inherently exhibit multi-scale features. To capture this, the PDE-Transformer augments the DiT backbone with a hierarchical, U-shaped architecture as seen in Figure 1. The model employs token downsampling and upsampling between transformer stages using PixelUnshuffle and PixelShuffle [Shi et al. 2016] layers, respectively. This design introduces an inductive bias similar to the classic U-Net [Ronneberger et al. 2015], allowing the model to efficiently process features at varying spatial resolutions while maintaining the scalability of transformer architectures.

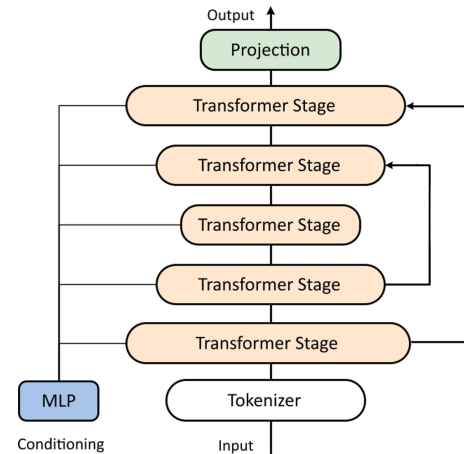


Fig. 1. Multiple Transformer stages combined in a U-Net type architecture [Holzschuh et al. 2025] with a MLP Conditioner.

3.2 Attention Mechanism: Shifted Window Attention

To mitigate the complexity of the global self-attention, the authors replace the global operation with Shifted Window Multi-Head Self-Attention [Liu et al. 2021]. This approach restricts the computation of self-attention to local, non-overlapping windows of tokens, significantly reducing the computational burden. It is important to prevent information from being isolated within these local patches and to ensure global propagation across the domain, therefore the

window partition is shifted by half the window size between consecutive layers (see Figure 2).

This allows the architecture to scale to larger domain sizes without the memory costs of global attention.

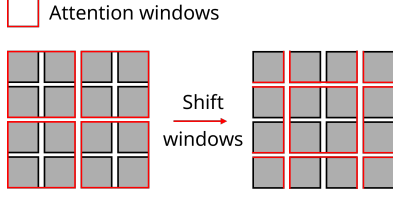


Fig. 2. Shifted windows for consecutive layers

3.3 Patching and Separate Channel

To process the grid data, the input is partitioned into patches of size $p \times p$, which are projected into token embeddings. To quantify the information capacity per token, the authors define the expansion rate $E(p) := d/(p^2T)$, where d is the embedding dimension and T is the temporal extent of the patch. In standard Mixed Channel (MC) approaches, all physical quantities (e.g., density, velocity) are compressed into a single token for a given patch. Consequently, as the number of channels increases, the expansion rate drops, leading to an overly compressed representation. To address this, the authors introduce a novel Separate Channel (SC) strategy (see Figure 3):

- **Decoupled Embeddings.** The SC strategy embeds each physical channel independently. This ensures that the expansion rate remains constant for each token, regardless of the number of physical variables involved, preventing information loss.
- **Axial Attention.** To allow these separated variables to interact, the model employs a channel-wise axial attention operation [Ho et al. 2019], enabling cross-channel communication without merging the tokens.

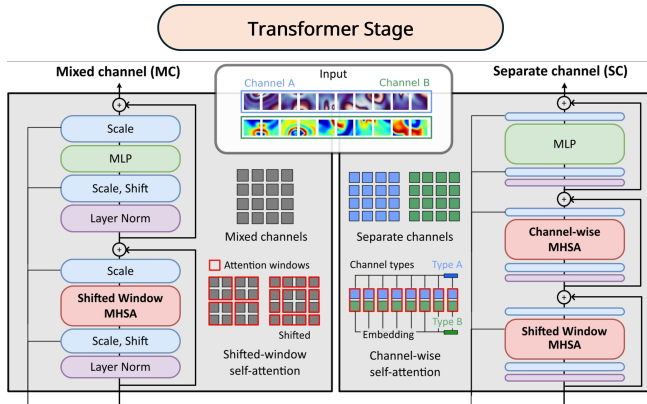


Fig. 3. Single Transformer Stage architecture [Holzschuh et al. 2025]. Comparison of Mixed Channel and Separate Channel embedding strategies

3.4 Conditioning and Boundary Conditions

The model incorporates conditioning mechanisms to handle varying simulation parameters and boundary types.

- **adaLN-Zero.** System parameters (e.g.: PDE type, diffusion time or channel type) are injected into the network using adaptive Layer Normalization (adaLN-Zero) blocks, which regress scale and shift factors for the normalized features.
- **Boundary Handling.** The architecture explicitly accounts for boundary topology. Periodic boundaries are implemented by cyclically rolling tokens during the window shift and non-periodic boundaries are handled by masking token interactions.

3.5 Training Modes: Supervised and Generative

A key feature of the PDE-Transformer is its versatility in training objectives, supporting both deterministic and probabilistic modeling:

- **Supervised Training.** For tasks with deterministic solutions (e.g., standard surrogate modeling), the network is trained to minimize the Mean Squared Error (MSE) between the prediction and the ground truth. The model functions as an autoregressive predictor, taking a sequence of T_p preceding snapshots to predict the state at the current time t , allowing for fast one-step inference:

$$u_{in} = [u_{t-T_p\Delta t}^S, \dots, u_{t-\Delta t}^S] \xrightarrow{\text{prediction}} [u_t^S]$$

- **Diffusion Training (Flow Matching).** For scenarios involving uncertainty or multi-modal distributions, the model employs Flow Matching [Lipman et al. 2023]. It learns a velocity field to transform a noise distribution into the data distribution via an Ordinary Differential Equation (ODE). This allows the model to sample from the full posterior distribution, providing uncertainty estimates.

4 EXPERIMENTAL EVALUATION AND SOTA COMPARISON

The performance of the PDE-Transformer was evaluated in two phases: pre-training on a set of 16 linear and non-linear PDEs and fine-tuning on challenging, out-of-distribution downstream tasks. The datasets are taken from APEBench [Koehler et al. 2024].

4.1 Pre-training Efficiency and Scaling

To ensure a fair evaluation, the PDE-Transformer was compared against representative baselines (DiT, UDiTs [Tian et al. 2024], U-Net, scOT [Herde et al. 2024], FactFormer [Li et al. 2023b]) using a "Small" (S) configuration, where all models possess a comparable number of trainable parameters and an embedding dimension of 96 elements.

Training Results. As summarized in Table 1, the PDE-S achieved the highest accuracy with a normalized Root Mean Square Error

(*nRMSE*) of 0.044, outperforming both the standard Diffusion Transformer (DiT-S, 0.066) and U-Net (0.075). Notably, PDE-S demonstrated superior efficiency, requiring only 7h 42m for training compared to 13h for DiT-S and 48h for U-Net. This increased speed is primarily due to the shifted window attention mechanism, which reduces computational complexity to $O(N)$, thereby keeping training times comparably low.

Table 1. Comparison of Model Performance and Training Costs (S Configuration) [Holzschuh et al. 2025]. The PDE-S achieves competitive accuracy with significantly lower training time compared to DiT and UDiT baselines.

Model	nRMSE ₁	nRMSE ₁₀	Time (h)	Params
DiT-S	0.066	0.78	13h 4m	39.8M
UDiT-S	0.042	0.39	18h 30m	58.9M
scOT-S	0.051	0.59	21h 11m	39.8M
FactFormer	0.069	0.65	12h 25m	3.8M
UNet	0.075	0.68	48h 00m	35.7M
PDE-S	0.044	0.36	7h 42m	33.2M

4.2 Generalization to Downstream Tasks

The model’s generalization capabilities were tested on three complex physics tasks: Active Matter, Rayleigh-Bénard Convection (RBC), and Shear Flow. These tasks introduce novel boundary conditions (e.g., walls) and geometries not present in the pre-training data.

Performance. As illustrated in Figure 4, the pre-trained PDE-S consistently outperformed state-of-the-art baselines trained from scratch in small configuration, including Fourier-Neural-Operator (FNO) [Li et al. 2020], Galerkin Transformer [Cao 2021], and OFormer [Li et al. 2023a]. It achieved an average nRMSE improvement of 42% over the second-best model (FNO).

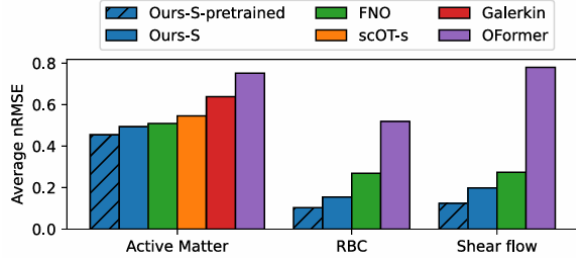


Fig. 4. Average rollout nRMSE of different models on downstream tasks [Holzschuh et al. 2025]. The pre-trained PDE-Transformer demonstrates superior stability and accuracy compared to baselines across all tasks.

Impact of Separate Channels. The experiments highlight the critical role of the SC embedding strategy for transfer learning, as shown in Table 2. Even though the MC and SC variants perform similarly when trained from scratch, the SC variant significantly outperforms MC when using pre-trained weights. Specifically, the SC approach yielded performance improvements ranging from 2.7× to

4.4× compared to MC on downstream tasks. This further reinforces the idea that separating physical variables enables the network to preserve its pre-trained knowledge more effectively, without the need for extensive training or fine-tuning of the PDE-specific layers.

Table 2. Comparison of PDE-Transformers using separate- and mixed-channel representations (S configuration) [Holzschuh et al. 2025]. Performance is evaluated by the mean nRMSE over the first 20 rollout steps.

Task	Mode	Scratch	Pre-trained	Improvement
Active Matter	SC	0.494	0.455	7.89%
	MC	0.493	0.479	2.84%
RBC	SC	0.155	0.104	32.90%
	MC	0.147	0.130	11.56%
Shear Flow	SC	0.199	0.125	37.19%
	MC	0.178	0.163	8.43%

5 EXPERIMENT: ROBUSTNESS TO NOISE INJECTION IN KOLMOGOROV FLOW

While foundation models excel at reproducing dynamics from clean data, their robustness against the inherent noise and sparsity of real-world observational data remains a critical factor for deployment in engineering scenarios. To evaluate this capability, an independent experiment was conducted using the Kolmogorov Flow dataset, specifically testing the pre-trained PDE-Transformer’s ability to act as a filter for unphysical noise and to maintain stability under perturbed initial conditions. The code is available under https://gitlab.lrz.de/SametKocbay/deeplearninginphysics_pdetransformer_experiment.git.

5.1 Methodology

The pre-trained PDE-Transformer (MC-Small configuration) was utilized to model 2D Kolmogorov Flow, a chaotic system governed by the Navier-Stokes equations. To simulate measurement error, Gaussian noise was injected into the initial state ($t = 0$) with intensities σ_{noise} varying from 5% to 50% relative to the input standard deviation. The model was trained to predict 29 future timesteps and evaluated using autoregressive rollouts over 29 timesteps.

Performance was evaluated using physics-informed diagnostics rather than solely relying on pixel-wise error, which can be misleading in chaotic systems where trajectories naturally diverge.

- **Enstrophy (ϵ).** Defined as $\epsilon(t) = \frac{1}{2} \int_{\Omega} \omega^2 dx$, measuring turbulence intensity and dissipation.
- **Energy Spectrum Analysis.** Analyzing the energy distribution $E(\kappa)$ in Fourier space to verify the restoration of the 2d energy cascade characteristic of 2D turbulence [Paret and Tabeing 1998].

5.2 Results

Energy Spectrum Analysis. As illustrated in Figure 5, the injection of Gaussian noise results in a flattened energy spectrum (red curve), characterized by unphysical energy elevation at high

wavenumbers. In contrast, the spectral analysis of the model’s output (cyan curve) demonstrates that the PDE-Transformer effectively filters this high-frequency noise within a single inference step. The model successfully restores the characteristic slope of the 2d energy cascade, aligning the predicted state closely with the ground truth physics.

While this suggests a strong learned inductive bias towards valid solutions of the Navier-Stokes equations, it is important to consider alternative explanations. This filtering behavior could also be attributed to the well-known spectral bias of neural networks [Rahaman et al. 2019] or the inherent blurring effect of the MSE loss function used during training, which tends to suppress high-frequency components regardless of their physical validity.

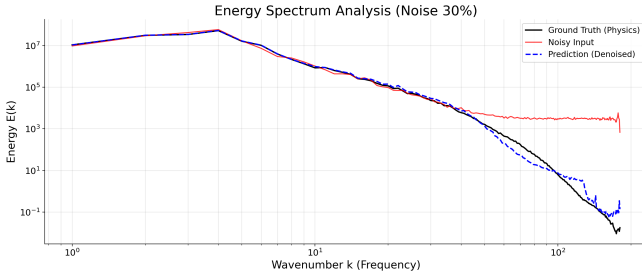


Fig. 5. Energy Spectrum Analysis. Comparison of kinetic energy distribution across spatial frequencies for the perturbed initial state (red), the model’s one-step prediction (cyan), and the ground truth (white). The prediction exhibits a restored spectral slope consistent with the 2d energy cascade.

Stability and Error Evolution. The analysis of predictive error and physical stability reveals a robust response to noise. As shown in Table 3, the nRMSE naturally scales with the intensity of the injected noise. At $t = 1$, the error for the 50% noise case (0.0385) is more than three times that of the baseline. However, as the rollout progresses to $t = 29$, the relative degradation decreases. The error for the 50% case (0.1922) is only marginally higher than the clean baseline (0.1557). This suggests that the model does not catastrophically diverge but rather settles into a valid, albeit shifted trajectory.

Table 3. nRMSE growth over time for varying noise levels at selected unroll steps 1, 15, 29.

σ_{noise}	nRMSE ₁	nRMSE ₁₅	nRMSE ₂₉
0%	0.0115	0.0937	0.1557
5%	0.0119	0.0940	0.1567
30%	0.0211	0.0945	0.1501
50%	0.0385	0.1140	0.1922

This physical stability is further evaluated through enstrophy analysis. As shown in Figure 6, the temporal evolution of enstrophy remains bounded throughout the training horizon (29 steps), even under high-magnitude noise injection (50%). The predictions accurately reproduce the expected oscillatory decay of the vorticity field, suggesting that the model preserves the underlying dissipative dynamics of the system rather than diverging into unphysical

states. However, extending the rollout beyond 29 steps reveals a divergence characterized by erratic behavior. This suggests that the accumulation of errors eventually drives the system state sufficiently out-of-distribution, compromising the model’s ability to maintain stable development.

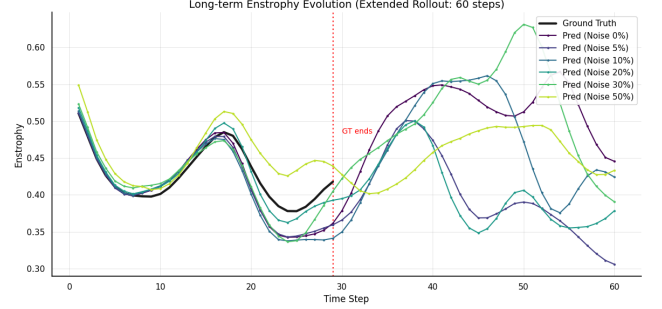


Fig. 6. Enstrophy evolution $\epsilon(t)$ over 60 unrolling steps for various noise levels. The ground truth (solid black line) shows the characteristic oscillatory decay of the enstrophy.

5.3 Evaluation of Noise Robustness

The experiments indicate that the PDE-Transformer possesses inherent robustness, effectively acting as a spectral filter to recover the characteristic 2d energy cascade even from significant noise contamination. It remains an open question whether this filtering capability stems from a learned physical inductive bias or is primarily a result of the spectral bias inherent to neural networks and the smoothing effect of the MSE loss function.

Despite this ambiguity regarding the mechanism, the functional outcome is physically stable within the training horizon. The model accurately reproduces the oscillatory decay of enstrophy for the first 29 steps, preserving the system’s dissipative dynamics. However, the observed divergence in enstrophy during extended rollouts reveals a limitation in long-term conservation. Future work should address this by incorporating physics-informed loss functions, such as explicit enstrophy conservation terms, to better constrain the solution space during fine-tuning.

6 CONCLUSION

The PDE-Transformer addresses the computational bottlenecks of DiTs in physics simulations by combining a multi-scale U-shaped architecture with shifted window attention. The proposed SC embedding strategy effectively decouples physical variables, leading to significant performance gains in downstream tasks compared to the MC approach. While the model demonstrates superior accuracy and efficiency against state-of-the-art baselines like FNO and scOT on regular 2D grids, its Diffusion capabilities need to be evaluated. Further, the extension to 3D grids plays a crucial role.

To overcome these geometric constraints, future work could be aimed to extend the architecture to 3D simulations and potentially integrate Graph Neural Operators to map irregular meshes into the transformer’s latent grid.

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